Zomato Restaurant Analysis

Bangalore restaurants: 5 business questions answered with SQL, Python and Power BI

Item	Details
Dataset	Zomato Bangalore restaurants, 51,717 rows x 17 columns, 0 duplicate rows
Tools	SQLite (DB Browser), Python (Pandas, Matplotlib, VS Code Jupyter), Power BI
Rating cleaning	Ratings stored as text (e.g. 4.1/5). 7,775 missing (15.03%) plus non-numeric values such as NEW. After conversion, 41,590 valid numeric ratings in the Power BI export.
Power BI dataset	51,609 rows x 12 columns (exact duplicates removed after cleaning)
Dashboard KPIs	Avg rating 3.70 | Avg cost for two Rs. 555.17 | 51,609 restaurant records

Summary of answers

#	Question	Tool	Answer in one line
1	Highest rated cuisines	SQL + Python	Pizza, Cafe, Italian (4.41) is the top cuisine combination; Modern Indian and Malaysian (4.31) lead as single cuisines.
2	Locations with most restaurants and avg cost	SQL	BTM leads with 5,098 restaurants (avg cost 396.48); Koramangala 5th Block is the costliest of the top locations (663.66).
3	Online order vs rating	SQL	Online = Yes averages 3.72 vs 3.66 for No. A small gap.
4	Restaurant types and ratings	Python	Quick Bites is the largest type (19,132, rating 3.55); Cafe rates higher (3.85) with 3,732 records.
5	Best value locations	Python + Power BI	City Market (1.09) and South Bangalore (1.08) give the highest rating per 100 rupees; BTM is lowest of the high-volume areas (0.85).

Q1. Which cuisines have the highest average rating?

Method. SQL groups the full cuisines string (cuisine combinations), keeps groups with at least 50 rated restaurants, and averages the numeric rating. Python splits the string into individual cuisines and applies the same minimum of 50 rated records. Both views are shown because they answer slightly different questions.

SQL result: top cuisine combinations (min 50 restaurants)

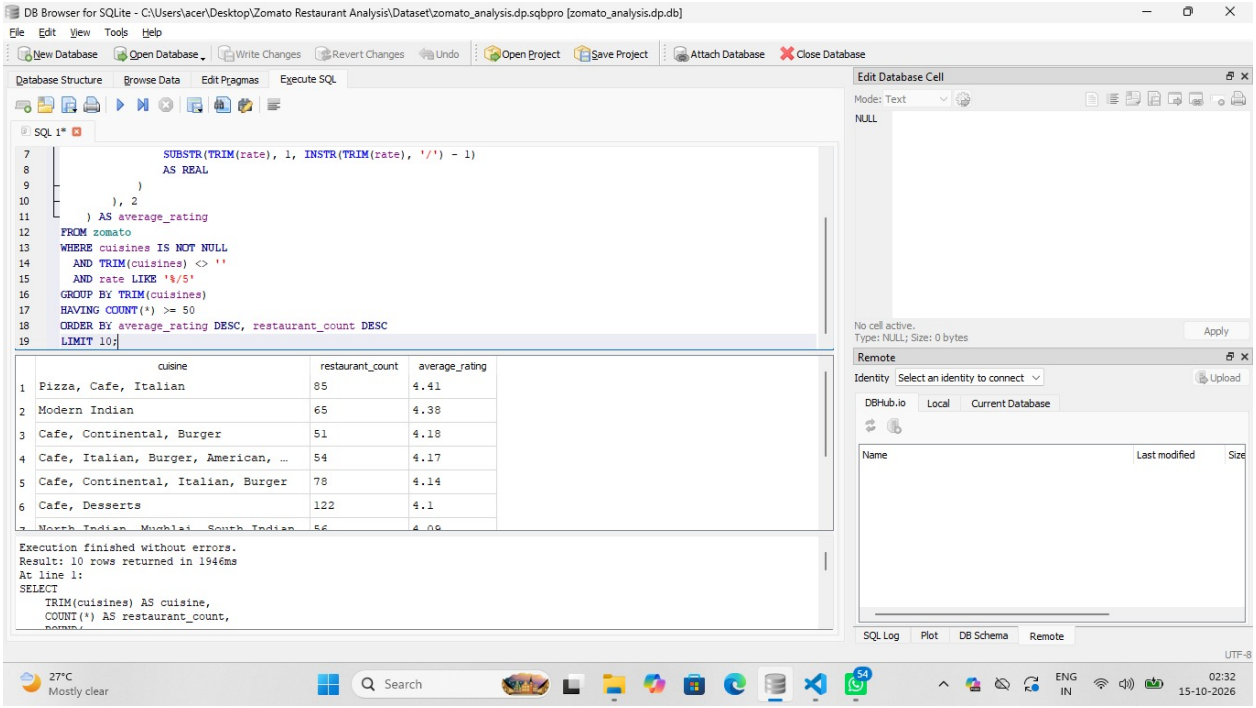
Cuisine combination	Restaurants	Avg rating
Pizza, Cafe, Italian	85	4.41
Modern Indian	65	4.38
Cafe, Continental, Burger	51	4.18
Cafe, Continental, Italian, Burger	78	4.14
Cafe, Desserts	122	4.10

Python result: top individual cuisines (min 50 rated records)

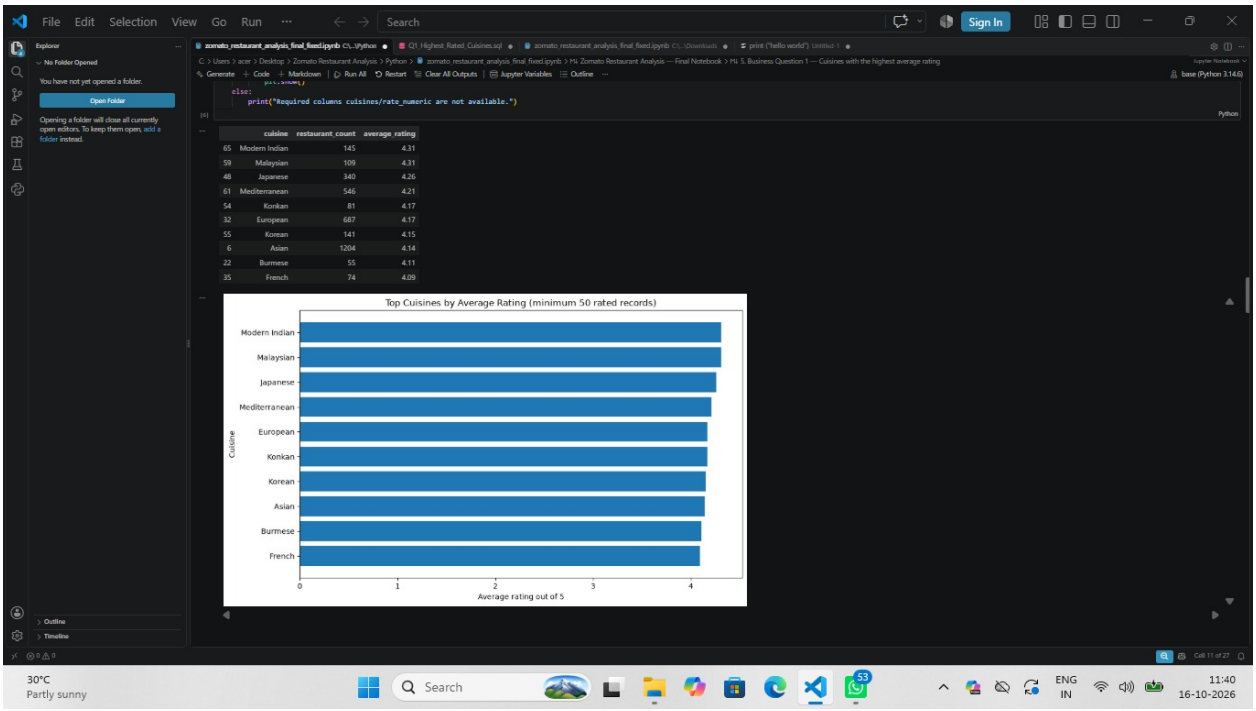
Cuisine	Records	Avg rating
Modern Indian	145	4.31
Malaysian	109	4.31
Japanese	340	4.26
Mediterranean	546	4.21
Konkan	81	4.17
European	687	4.17
Korean	141	4.15
Asian	1,204	4.14
Burmese	55	4.11
French	74	4.09

Insight. Cafe-style combinations and Modern Indian rate highest among combinations, while Modern Indian, Malaysian and Japanese lead among single cuisines. Asian (1,204 records at 4.14) is the most reliable high-rating cuisine because of its large sample.

Q1 evidence: screenshots



SQL screenshot: Q1 query and output (DB Browser for SQLite)



Python screenshot: top cuisines by average rating

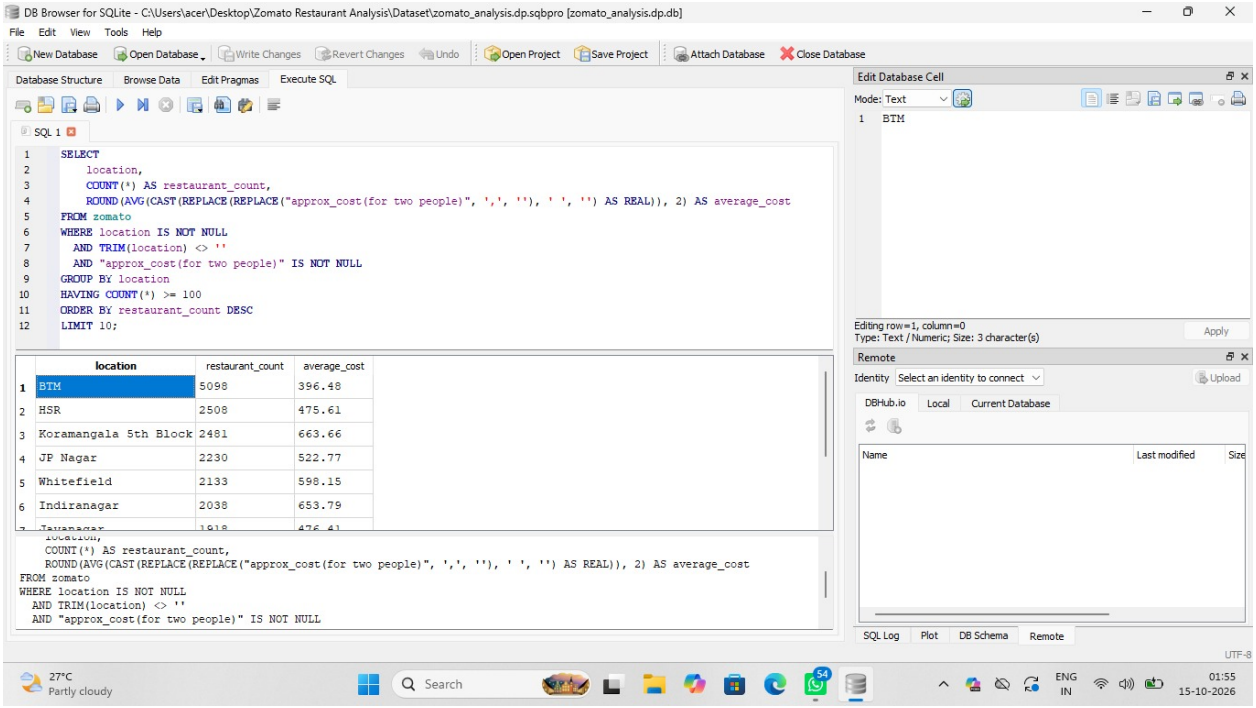
Q2. Which locations have the most restaurants and what is the average cost for two?

Method. SQL groups by location, counts restaurants that have a cost value, averages approx_cost(for two people), and keeps locations with at least 100 restaurants. Python counts all records per location, so counts are slightly higher (for example BTM 5,124 vs 5,098) because Python includes rows with missing cost. Average cost is identical in both.

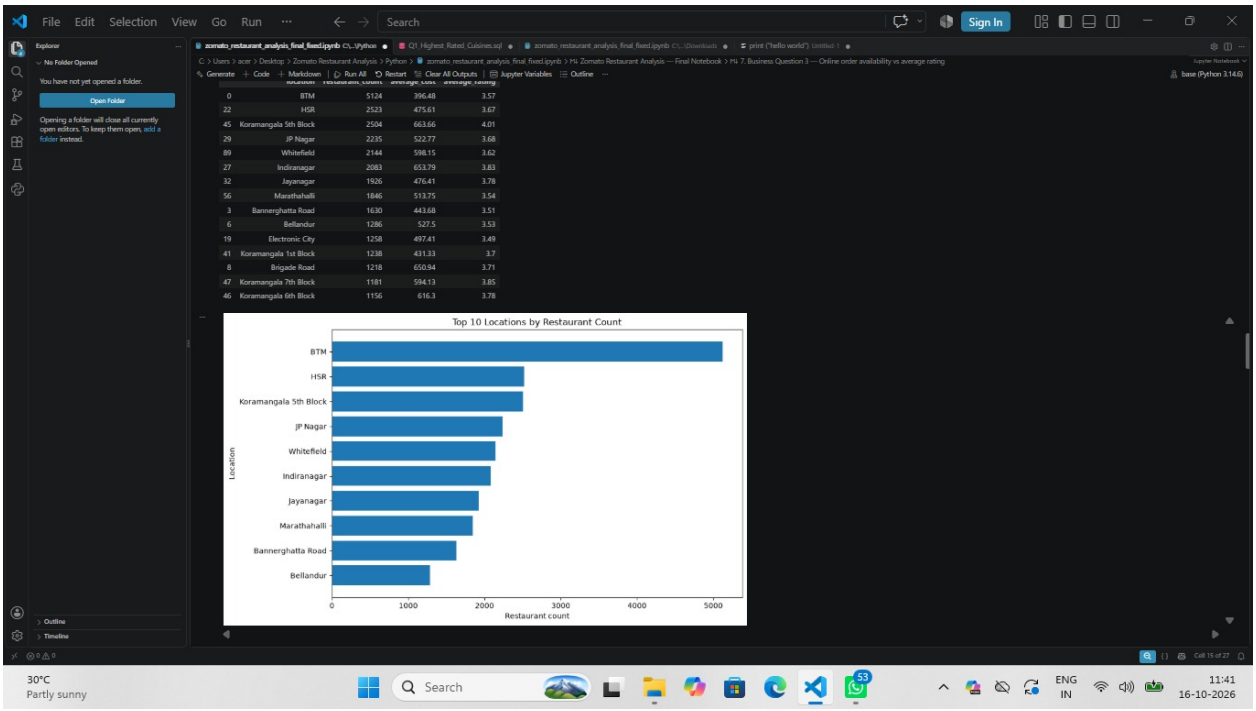
Location	SQL count (with cost)	Python count (all)	Avg cost for two (Rs.)	Avg rating (Python)
BTM	5,098	5,124	396.48	3.57
HSR	2,508	2,523	475.61	3.67
Koramangala 5th Block	2,481	2,504	663.66	4.01
JP Nagar	2,230	2,235	522.77	3.68
Whitefield	2,133	2,144	598.15	3.62
Indiranagar	2,038	2,083	653.79	3.83
Jayanagar	1,918	1,926	476.41	3.78
Marathahalli	-	1,846	513.75	3.54
Bannerghatta Road	-	1,630	443.68	3.51
Bellandur	-	1,286	527.50	3.53

Insight. BTM has almost double the restaurants of the next location (HSR), so it is the most competitive zone, but it is also one of the cheapest. Koramangala 5th Block and Indiranagar combine high cost (about Rs. 650+) with high ratings (4.01 and 3.83), a premium segment. Rows marked dash were outside the SQL top 7 shown in the screenshot.

Q2 evidence: screenshots



SQL screenshot: Q2 query and output



Python screenshot: top 10 locations by restaurant count

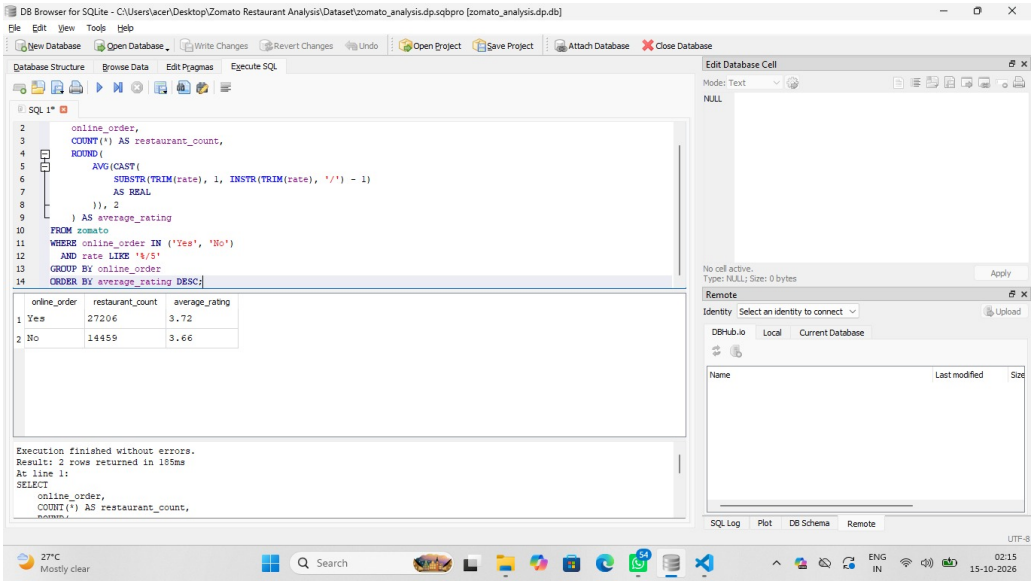
Q3. Do restaurants offering online orders have a higher rating?

Method. SQL groups by online_order and averages numeric ratings, using only restaurants with a valid rating (rate in the form x/5). Python gives the same results and also shows total records.

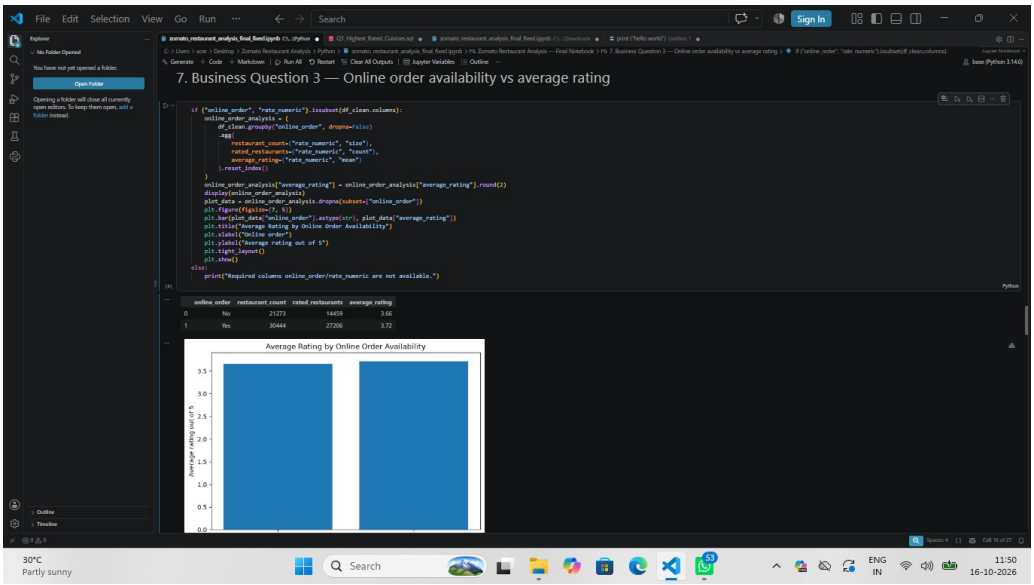
Online order	Total records	Rated restaurants	Avg rating
Yes	30,444	27,206	3.72
No	21,273	14,459	3.66

Insight. Restaurants with online ordering are rated 0.06 points higher on average. This is a small difference. It is an observed pattern only; it does not prove that online ordering causes better ratings, because location, price and restaurant type also affect ratings.

Bonus finding: table booking shows a much larger gap. Restaurants with Book Table = Yes average 4.14 (6,304 rated) versus 3.62 for No (35,361 rated), a difference of 0.52.



SQL screenshot: Q3 query and output



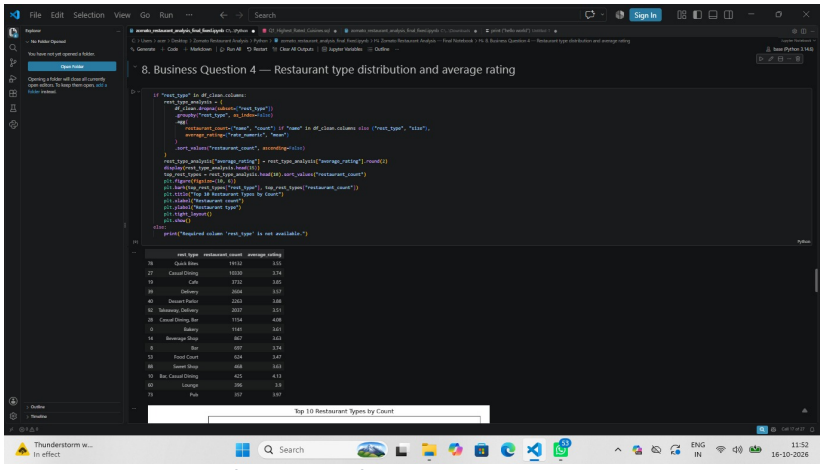
Python screenshot: average rating by online order availability

Q4. What is the distribution of restaurant types and their average ratings?

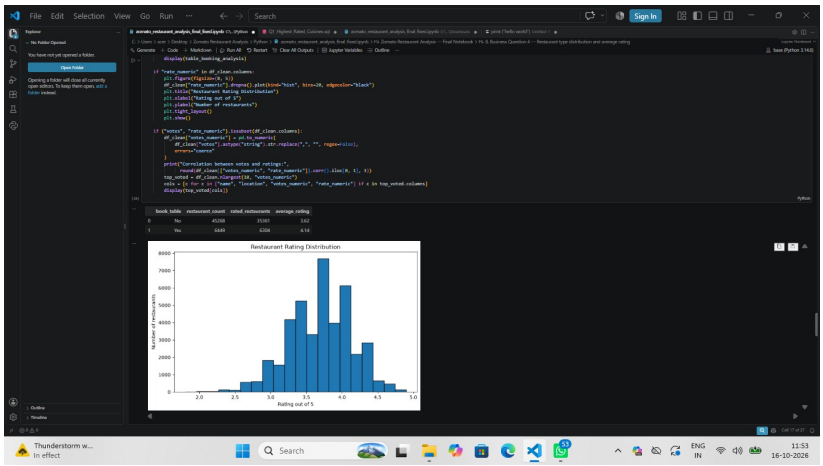
Method. Python groups by rest_type and calculates record count and average numeric rating, shown with a bar chart of the top 10 types by count.

Restaurant type	Records	Avg rating
Quick Bites	19,132	3.55
Casual Dining	10,330	3.74
Cafe	3,732	3.85
Delivery	2,604	3.57
Dessert Parlor	2,263	3.88
Takeaway, Delivery	2,037	3.51
Casual Dining, Bar	1,154	4.08
Bakery	1,141	3.61
Beverage Shop	867	3.63
Bar	697	3.74

Insight. Quick Bites dominates volume but has one of the lower ratings (3.55). Formats that combine dining with a bar rate highest (Casual Dining, Bar at 4.08; Bar, Casual Dining at 4.13 with 425 records). Cafes (3.85) and Dessert Parlors (3.88) also rate above the overall average. Frequent does not mean highest rated.



Python screenshot: restaurant type counts and average ratings



Python screenshot: rating distribution histogram (most ratings fall between 3.5 and 4.2)

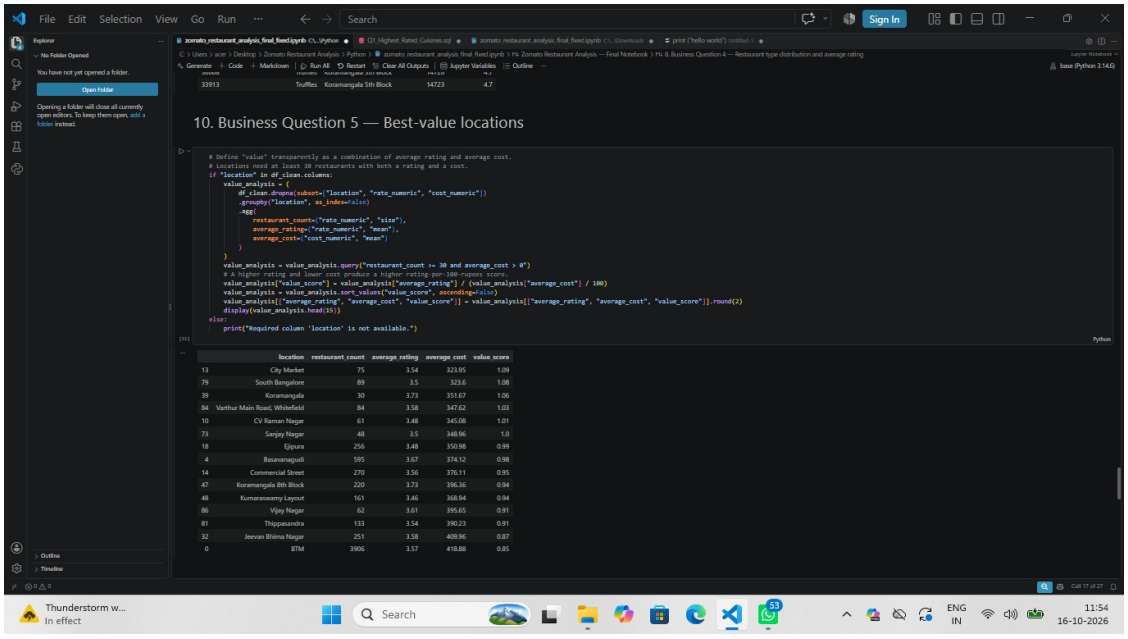
Q5. Which neighbourhoods offer the best value (high rating at low cost)?

Method. Value Score = Average Rating / (Average Cost / 100), i.e. rating points per 100 rupees spent. Only locations with at least 30 restaurants having both a rating and a cost are included. A higher score means better value.

Location	Restaurants	Avg rating	Avg cost (Rs.)	Value score
City Market	75	3.54	323.95	1.09
South Bangalore	89	3.50	323.60	1.08
Koramangala	30	3.73	351.67	1.06
Varthur Main Road, Whitefield	84	3.58	347.62	1.03
CV Raman Nagar	61	3.48	345.08	1.01
Sanjay Nagar	48	3.50	348.96	1.00
Ejipura	256	3.48	350.98	0.99
Basavanagudi	595	3.67	374.12	0.98
Commercial Street	270	3.56	376.11	0.95
Koramangala 8th Block	220	3.73	396.36	0.94
BTM (for comparison)	3,906	3.57	418.88	0.85

Insight. The best-value areas are the lower-cost neighbourhoods (about Rs. 320 to 350 for two) that still keep ratings around 3.5 to 3.7. Basavanagudi (595 restaurants) and Ejipura (256) are the strongest picks with a large sample. BTM has the most restaurants but a lower value score (0.85) because of its higher average cost of 418.88.

Power BI dashboard. Built from the cleaned CSV with four slicers (Location, Online Order, Book Table, Restaurant Type) and these visuals: Top 10 Locations by Restaurant Count, Average Cost by Location, Avg Rating by Location, Top 10 Best-Value Locations, a scatter chart of Average Rating vs Average Cost, and three KPI cards (Avg Rating 3.70, Avg Cost Rs. 555.17, Total Records 51,609).



Python screenshot: value score calculation for best-value locations

Key takeaways

- BTM is the most crowded restaurant location (5,098 to 5,124 records) with a below-average cost of about Rs. 396.
- Cafe-style and Modern Indian cuisines earn the highest average ratings; Asian cuisine is the strongest high-volume performer (4.14 over 1,204 records).
- Online ordering shows only a small rating advantage (3.72 vs 3.66); table booking shows a much larger one (4.14 vs 3.62).
- Quick Bites is the largest format but rates lower (3.55); bar and dining combinations rate highest (about 4.1).
- City Market, South Bangalore, Basavanagudi and Ejipura offer the best rating per rupee.

Notes on the data

- Average ratings use only restaurants with a valid numeric rating. About 15% of ratings are missing and some are labelled NEW.
- SQL ranks the cuisines column as stored (combinations); Python splits it into individual cuisines. Both are reported.
- SQL location counts include only rows with a cost value; Python counts include all rows.
- Results are descriptive patterns and do not prove cause and effect.
- Approximate cost for two is an estimate, not a final bill.

Project files

Folder	Contents
Sql/	Q1_Highest_Rated_Cuisines.sql, Q2_Location_Restaurant_Analysis.sql, Q3.Online_Order_vs_Rating.sql
Python/	zomato_restaurant_analysis_final_fixed.ipynb, zomato_powerbi_dashboard.cleaned.csv
PowerBi/	zomato_restaurant_analysis.pbix
Screenshots/	01_SQL, 02_Python, 03_PowerBI screenshots
Dataset/	Source: Kaggle, Zomato Bangalore Restaurants (himanshu7273)